

Partial Algebraic Shifting

Antony Della Vecchia

Joint work with M. Joswig and F. Lenzen

TU Berlin

2025-06-03



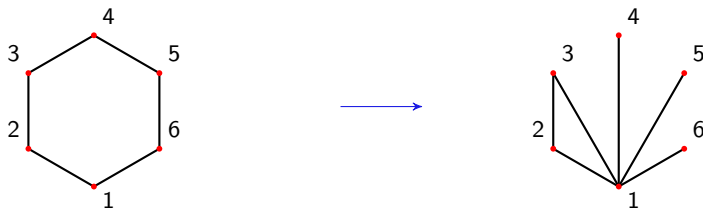
Algebraic Shifting

Definition

$C \in \binom{[n]}{k}$ is shifted if for any $\sigma \in C$, $\sigma|_i^j \in C$ for $i \in \sigma$ and $j < i$.

Theorem (Kalai '84, Björner-Kalai '88)

Given K a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(K)$ with the same f -vector and Betti numbers as K .



Exterior Algebraic Shifting

Definition

Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \mathrm{GL}(n, \mathbb{E})$.

- ▶ The k th compound matrix $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$ -matrix of k -minors of g .
- ▶ $g^{\wedge K_k}$ is the $|K_k| \times \binom{n}{k}$ -submatrix of $g^{\wedge k}$ with rows indexed by K_k .

Definition

The partial (exterior) shift of K by g is

$$\Delta_g^{\mathbb{F}}(K) := \bigcup_k \left\{ \sigma \in \binom{[n]}{k} : g_{*\sigma}^{\wedge K_k} \notin \mathrm{span}_{\mathbb{E}}((g_{*\rho}^{\wedge K_k})_{\rho <_{\mathrm{lex}} \sigma}) \right\}.$$



Small Example

Example

For $g = (x_{ij})_{1 \leq i, j \leq 4}$ and $K = \{\{1, 2\}, \{2, 3\}, \{4\}\}$, $\mathbb{F} = \text{GF}(2)$.

$$g^{\wedge K_2} = \begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{22}x_{13} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ x_{21}x_{32}+x_{22}x_{31} & x_{21}x_{33}+x_{23}x_{31} & x_{21}x_{34}+x_{24}x_{31} & x_{22}x_{33}+x_{23}x_{32} & x_{22}x_{34}+x_{24}x_{32} & x_{23}x_{34}+x_{24}x_{33} \end{pmatrix}.$$

Which row reduces to,

$$\begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{13}x_{22} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ 0 & x_{11}x_{21}x_{22}x_{33} & x_{11}x_{21}x_{22}x_{34} & x_{11}x_{22}^2x_{33} & x_{11}x_{22}^2x_{34} & x_{11}x_{22}x_{23}x_{34} \\ & +x_{11}x_{21}x_{23}x_{32} & +x_{11}x_{21}x_{24}x_{32} & +x_{11}x_{22}x_{23}x_{32} & +x_{11}x_{22}x_{24}x_{32} & +x_{11}x_{22}x_{24}x_{33} \\ & +x_{12}x_{21}^2x_{33} & +x_{12}x_{21}^2x_{34} & +x_{12}x_{21}x_{22}x_{33} & +x_{12}x_{21}x_{22}x_{34} & +x_{12}x_{21}x_{24}x_{33} \\ & +x_{13}x_{21}^2x_{32} & +x_{14}x_{21}^2x_{32} & +x_{13}x_{21}x_{22}x_{32} & +x_{14}x_{21}x_{22}x_{32} & +x_{13}x_{21}x_{24}x_{32} \\ & +x_{12}x_{21}x_{23}x_{31} & +x_{12}x_{21}x_{24}x_{31} & +x_{12}x_{22}x_{23}x_{31} & +x_{12}x_{22}x_{24}x_{31} & +x_{14}x_{21}x_{23}x_{32} \\ & +x_{13}x_{21}x_{22}x_{31} & +x_{14}x_{21}x_{22}x_{31} & +x_{13}x_{22}^2x_{31} & +x_{14}x_{22}^2x_{31} & +x_{13}x_{22}x_{24}x_{31} \\ & & & & & +x_{14}x_{22}x_{23}x_{31} \end{pmatrix}.$$

$$\Delta(K) = \{\{1, 2\}, \{1, 3\}, \{4\}\}$$



Algebraic Exterior Shifting

Generic Initial Ideals

For $\sigma = \{\sigma_1 < \dots < \sigma_k\}$ set $e_\sigma = e_{\sigma_1} \wedge \dots \wedge e_{\sigma_k}$

$$I_K := (e_\sigma : \sigma \notin K) \subset \bigwedge \mathbb{F}^n$$

The exterior face ring is,

$$\mathbb{F}[K] := \bigwedge \mathbb{F}^n / I_K,$$

and

$$\mathbb{F}[\Delta(K)] = \bigwedge \mathbb{F}^n / \text{gin}(I_K)$$

See [Herzog-Tarai '99] Stable Properties of algebraic shifting.



Partial Algebraic Shifting

Consider now $\{\tau(w) \in \mathrm{GL}(n, \mathbb{E}) : w \in S_n\}$.

Let $c_n := (1 \dots n) \in S_n$.

Theorem (DV-Joswig-Lenzen '24+)

For any simplicial complex K and $w \in S_n$ such that $c_n \leq w$ in the weak Bruhat order we have the following,

1. $\Delta_{\tau(w)}(K)$ is homotopy equivalent to a wedge of spheres.
2. $\Delta_{\tau(w)}(K)$ has the same Betti numbers as K .



Real Projective Plane

$$K = \mathbb{RP}_6^2, \mathbb{F} = \mathbb{Q}$$

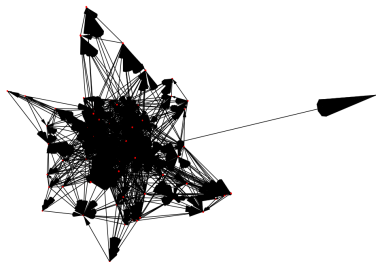


Figure: The partial shift graph starting from $\mathbb{RP}_6^2$, generated using OSCAR.



LU decomposition

Lemma

$\Delta_{gb}(K) = \Delta_g(K)$ for any upper triangular b and any simplicial complex K .

LU decomposition For $g \in GL(n, \mathbb{E})$ with non zero leading principal minors, there exists l lower triangular, u upper triangular such that

$$g = lu$$

Example

$K = \{\{1, 2\}, \{2, 3\}, \{4\}\}$, and $I^{\wedge K_2}$ reduces to

$$\begin{pmatrix} x_{13}x_{24}+x_{14}x_{23} & x_{12}x_{24}+x_{14} & x_{24} & x_{12}x_{23}+x_{13} & x_{23} & 1 \\ 0 & x_{12}x_{23}x_{24}x_{34}+x_{12}x_{24}^2 & x_{23}x_{24}x_{34} & x_{12}x_{23}^2x_{34}+x_{12}x_{23}x_{24} & x_{23}^2x_{34} & x_{23}x_{34} \\ & +x_{13}x_{24}x_{34}+x_{14}x_{24} & +x_{24}^2 & +x_{13}x_{23}x_{34}+x_{14}x_{23} & +x_{23}x_{24} & +x_{24} \end{pmatrix},$$



Bruhat Decomposition

Bruhat Decomposition Let $B \subset GL(n, \mathbb{E})$ be the subgroup of upper triangular matrices,

$$GL(n, \mathbb{E}) = \coprod_{w \in S_n} BwB .$$

$$w_0 = (1 \ n)(2 \ n-1) \dots (\lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} \rceil + 1)$$



Bruhat Decomposition

Bruhat Decomposition Let $B \subset \mathrm{GL}(n, \mathbb{E})$ be the subgroup of upper triangular matrices,

$$\mathrm{GL}(n, \mathbb{E}) = \coprod_{w \in S_n} BwB .$$

$$w_0 = (1 \ n)(2 \ n-1) \dots (\lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} \rceil + 1)$$

If g is generic, then so is w_0g , hence

$$\Delta_{w_0g}(K) = \Delta_g(K) = \Delta(K).$$

For the same reason w_0g there exist some lower triangular l , and upper triangular u' , u such that,

$$\begin{aligned} w_0g &= lu = w_0u'w_0u \\ g &= u'w_0u \end{aligned}$$

$g \in Bw_0B$, and $\Delta_{u'w_0}(K) = \Delta(K)$



Inversions

Definition

For $w \in S_n$, let

$$\text{Inv } w := \{(i, j) : i < j, i \cdot w > j \cdot w\}$$

$$\ell(w) := |\text{Inv } w|$$

$$w_0 = \operatorname{argmax}_{w \in S_n} \ell(w)$$



Inversions

Definition

For $w \in S_n$, let

$$\text{Inv } w := \{(i, j) : i < j, i \cdot w > j \cdot w\}$$

$$\ell(w) := |\text{Inv } w|$$

$$w_0 = \operatorname{argmax}_{w \in S_n} \ell(w)$$

Example

Take $w = (2 \ 3)$, $\text{Inv } w = \{(1, 2), (1, 3)\}$.

$$\begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & y & x \\ & 1 & 0 \\ & & 1 \end{pmatrix}.$$



Inversions

Definition

For $w \in S_n$, let

$$\text{Inv } w := \{(i, j) : i < j, i \cdot w > j \cdot w\}$$

$$\ell(w) := |\text{Inv } w|$$

$$w_0 = \operatorname{argmax}_{w \in S_n} \ell(w)$$

Example

Take $w = (2 \ 3)$, $\text{Inv } w = \{(1, 2), (1, 3)\}$.

$$\begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & y & x \\ & 1 & 0 \\ & & 1 \end{pmatrix}.$$

Minimal representatives

$$u(w) := \begin{pmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & \ddots & & \vdots \\ & & 1 & u_{n-1,n} \\ & & & 1 \end{pmatrix} \quad \text{with} \quad u_{ij} = \begin{cases} x_{ij} & \text{if } (i, j) \in \text{Inv } w, \\ 0 & \text{otherwise} \end{cases}.$$

We call $\tau(w) := u(w)w$ the Rothe matrix.



Weak Bruhat order

Definition (Björner-Brenti '05)

The weak Bruhat right order is a partial ordering on S_n ,

$$w = uv \geq u$$

if $\ell(w) = \ell(u) + \ell(v)$.

Lemma

For $u, v \in S_n$ with $uv \geq u$ if and only if

$$\text{Inv } uv = \text{Inv } u \cup (\text{Inv } v) \cdot u^{-1}$$

$$\emptyset = \text{Inv } u \cap \text{Inv } v$$



Rothe Matrix of the n cycle

Consider $c_n = (1\ 2 \dots n)$,

- ▶ For $1 \leq i < j < n$, $i \cdot c_n < j \cdot c_n$
- ▶ $\text{Inv } c_n = \{(i, n) : 1 \leq i < n\}$

$$\tau(c_n) = \begin{pmatrix} x_{1,n} & 1 & & \\ \vdots & & \ddots & \\ x_{n-1,n} & & & 1 \\ 1 & 0 & \dots & 0 \end{pmatrix}.$$

For $w \geq c_n$

- ▶ $\text{Inv } c_n \subset \text{Inv } w$
- ▶ $\tau(c_n)$ and $\tau(w)$ have the same first column.



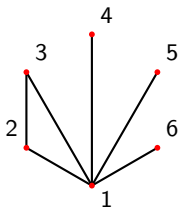
Near cones

Definition

A simplicial complex K is a near cone if for every $\sigma \in K$ with $1 \notin \sigma$ we have $\sigma|_i^1 \in K$ for each $i \in \sigma$.

Lemma (Björner-Kalai '88)

Let K be a near cone. Then K is homotopy equivalent to a wedge of spheres and $\beta_k(K) = |\{\sigma \in K^k : \sigma \cup \{1\} \notin K\}|$.



Permutation Lemma

Lemma (Björner-Kalai '88)

Let $g \in \text{GL}(n, \mathbb{E})$ be generic and suppose for $\tau \in \binom{[n]}{k}$ there exists $\gamma \in \mathbb{E}$ such that,

$$g_{\tau}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma_{\sigma} g_{\sigma}^{\wedge K_k}.$$

If $\pi \in S_n$, then there exists $\gamma' \in \mathbb{E}$ such that

$$g_{\tau \cdot \pi}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma'_{\sigma \cdot \pi} g_{\sigma \cdot \pi}^{\wedge K_k}$$



Shifting to near cones

Lemma (DV-Joswig-Lenzen '24+)

Let K be a simplicial complex on n vertices. Further, let $g \in \mathrm{GL}(n, \mathbb{E})$ be a matrix whose entries in the first column are algebraically independent over $\mathbb{F}$ and the other entries of g . Then $\Delta_g(K)$ is a near cone.

Proof sketch

Let $\pi = (1 \ i)$ and suppose $\tau \notin K$,

$$g_{\tau}^{\wedge K_k} = \sum_{\sigma < \tau} \gamma_{\sigma} g_{\sigma}^{\wedge K_k}.$$

Then there exists $\gamma' \in \mathbb{E}$ such that,

$$g_{\tau \cdot \pi}^{\wedge K_k} = \sum_{\sigma < \tau \cdot \pi} \gamma'_{\sigma} g_{\sigma}^{\wedge K_k}$$



Preserving Betti Numbers

Lemma (DV-Joswig-Lenzen '24+)

Let $v, w \in S_n$ with $\ell(vw) = \ell(v) + \ell(w)$ then $\tau(vw)^{K_k}$ and $(\tau(v)\tau(w)^v)^{\wedge K_k}$ have the same column matroid.

Proof sketch of shifting by $w \geq c_n$ preserves Betti numbers

It is enough to show,

$$|\{\sigma \in \Delta_{\tau(w_0)}(K^k) : 1 \in \sigma\}| = |\{\sigma \in \Delta_{\tau(w)}(K^k) : 1 \in \sigma\}|.$$



Preserving Betti Numbers

Lemma (DV-Joswig-Lenzen '24+)

Let $v, w \in S_n$ with $\ell(vw) = \ell(v) + \ell(w)$ then $\tau(vw)^{K_k}$ and $(\tau(v)\tau(w)^v)^{\wedge K_k}$ have the same column matroid.

Proof sketch of shifting by $w \geq c_n$ preserves Betti numbers

It is enough to show,

$$|\{\sigma \in \Delta_{\tau(w_0)}(K^k) : 1 \in \sigma\}| = |\{\sigma \in \Delta_{\tau(w)}(K^k) : 1 \in \sigma\}|.$$

Use lemma and do some algebra in the subscripts of $\Delta_{\tau(w)}$ and $\Delta_{\tau(w_0)}$



Problem How and or when can we find $\Delta(K)$ from $\Delta_{\tau(c_n)}(K)$?



Problem How and or when can we find $\Delta(K)$ from $\Delta_{\tau(c_n)}(K)$?



<https://arxiv.org/abs/2410.24044>

Thank You!

